

LEDGER · 2026-08-29 · AGRICULTURAL BIOTECHNOLOGY

High-Purity L-Amino Acid Biostimulant via Keratinase-Driven Feather Hydrolysis

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Enzymatic plant/soy amino acids
Entry application	Foliar biostimulant application for high-value organic horticulture
Measured advantage	1.0x parity on bioavailable L-amino acid purity and ~1.5x on proline content versus enzymatic soy amino acid incumbent
Time to first revenue	6 months
Gross margin at parity	32.5%
Startup CapEx	\$138,000
Payback from first sale	44.1 months
Capital productivity	1.08x

Why it failed

- Voided by: Economics undisclosed, Negative control / sub-2x, Form-factor / handling regression
- Below the 2x step-change bar
- Margin fails the operating-cost check

Assessment

The venture explicitly concedes failure against the real market leader (enzymatic soy hydrolysates), achieving only 1.0x parity on bioavailable L-amino acid purity and 1.5x on proline content. Gross margin is severely depressed at ~32.6% due to high energy costs from spray drying. Additionally, the fine hygroscopic powder presentation introduces form factor handling regression compared to liquid incumbents.

Also flagged

- Duplicate references inflating the evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-29 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.